

THE UNIVERSITY OF NEW SOUTH WALES
SCHOOL OF PHYSICS
THIRD YEAR LABORATORY

PROPERTIES OF LASER LIGHT

PLL

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Properties of Laser Light

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 - extracts from "Optics Guide 4", Melles Griot (1988).

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 C.C. Eaglesfield, "Laser Light" (1987) pp 179-182

3. **Polarization and intensity fluctuations in He-Ne lasers**
 E. Fortin and D. Singh, Am. J. Phys. **49** 891 (1981).

HOW DOES A HELIUM NEON LASER WORK?

All lasers depend for their operation on the phenomenon of stimulated emission of radiation. For this to occur, it is necessary for some atoms to exist in an excited energy state. In an excited atom an electron may decay to a lower energy level, emitting a photon of energy $E = h\nu$, where h is Planck's constant and ν is the photon frequency. This is spontaneous emission of radiation, or luminescence. If the emitted photon passes close to another atom in a similar excited state, it may induce an identical transition. In this case the emitted and stimulating photons will be identical in every respect, having the same frequency, phase, direction, and polarization — they will be spatially and temporally coherent. This is stimulated emission of radiation.

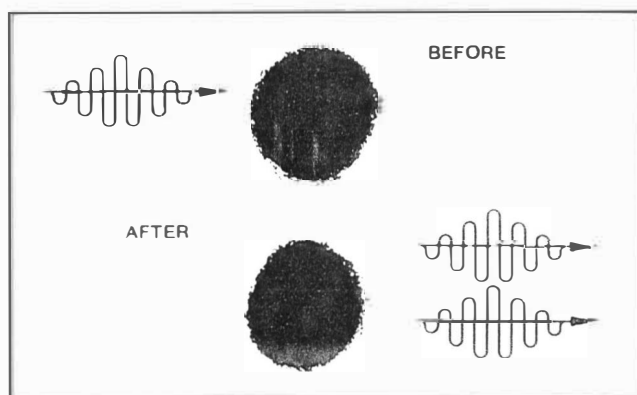
To achieve laser oscillation it is necessary to make the probability of stimulated emission exceed that of spontaneous emission and absorption. This is done by creating a population inversion, in which more atoms exist in an excited energy state than in some lower state to which direct radiative transition is possible. This situation must be maintained or laser action will cease.

In the helium neon system, a mixture of helium and neon gases is contained in a narrow bore tube. These gases are replenished as needed from a reservoir. The mixture is predominantly helium, with about 10% neon. Various isotopic mixtures have been used to improve the power output and stability of lasers. Melles Griot helium neon lasers are filled with a proprietary mixture which has the effect of

maximizing the ratio of beam power to plasma volume. An electrical discharge ionizes the gas inside the narrow bore. The plasma so formed contains many helium atoms in metastable states. In these states one of the valence electrons has been raised from the 1s and 2s level, where it will remain for a long time, the downward transition being forbidden by the selection rules of quantum mechanics. Neon has an atomic number of 10, and has 10 electrons normally residing in the $1s^2 2s^2 2p^6$ configuration. Collisions between metastable helium atoms and neon atoms cause one of the neon electrons in the 2p level to be raised to the 4s or 5s levels. The energy match between the 2s helium levels and the 4s and 5s neon levels is very close. It is therefore much more probable that this excitation will occur than one to the 3s or 3p levels. On exciting a neon atom in this way, the helium atom drops back to the ground state. However, because there are many more helium atoms than neon, the metastable collision probability remains high. For neon laser action to occur, it is necessary to maintain a greater population of neon atoms with electrons in the 4s and 5s levels than in the 3p level. This condition is satisfied by the selective excitation process just described, and by the low natural population of the 3p levels. A number of alternative downward transitions are available to the excited neon electrons. The major neon laser line transitions are indicated on the energy level diagram.

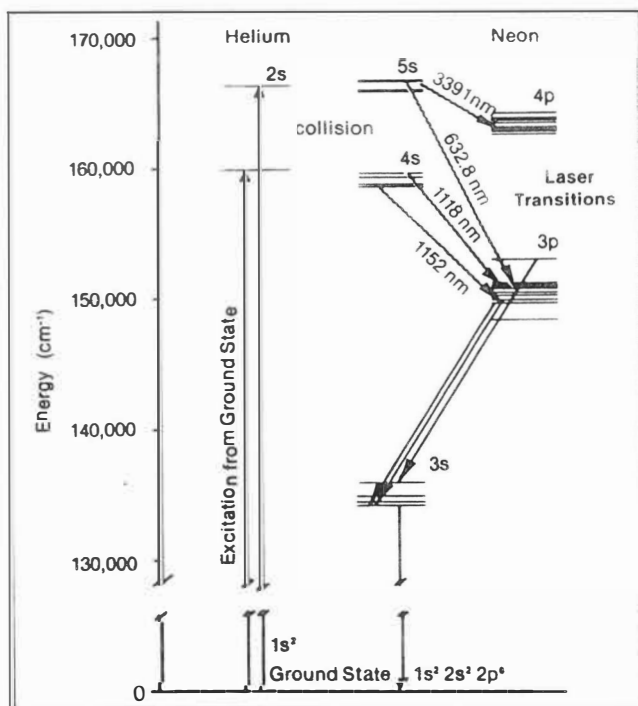
Some of the radiation emitted will be in a direction approximately parallel to the tube bore. The rest will be emitted through the tube walls and does not contribute to the laser operation. For a photon traveling along the bore, there is a chance of passage close to a second excited neon atom and stimulation of the emission of another identical photon. The probability of this occurrence depends on the gas density and length of the bore. For a single pass along the bore the probability is on the order of 10%.

Mirrors are placed at either end of the tube, reflecting photons many times along the length of the bore. In this way the likelihood of stimulated emission is further increased. Each stimulated photon is capable of stimulating another, and each is in perfect coherence with those already present. In this way, light amplification is achieved.

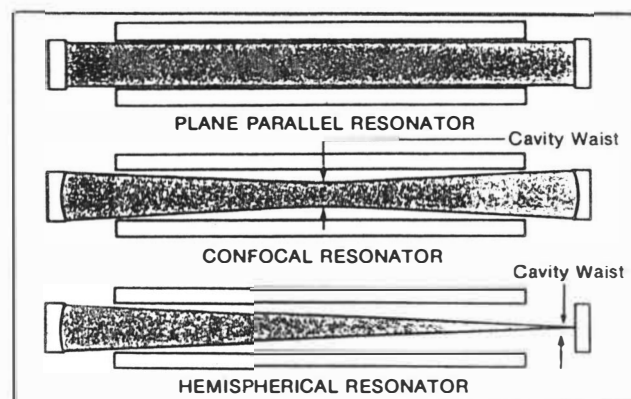


STIMULATED EMISSION, an artist's conception.

The two mirrors form a resonant optical cavity, and it is this cavity which is responsible for the selective gain characteristics of the laser. Cavity mirrors are coated to have a very high reflectance at the laser wavelength of interest, but to transmit unwanted laser lines. By retaining only the desired wavelength photons in the bore, amplification is restricted to only this wavelength. Similarly, misdirected radiation of the desired wavelength escapes from the cavity and is not amplified, giving rise to the highly directional nature of the laser output. In order to make use of the desired laser radiation, some of it must be allowed to escape from the cavity as a beam. In practice, one of the mirrors is coated to permit about 1% transmission at the laser wavelength. It is this 1% of light which escapes that forms the usable output of the laser. It follows that the internal beam is 100 times more powerful than the output beam.



ENERGY LEVEL DIAGRAMS for helium and neon. Levels are identified in spectroscopic notation, and major laser transitions are shown.

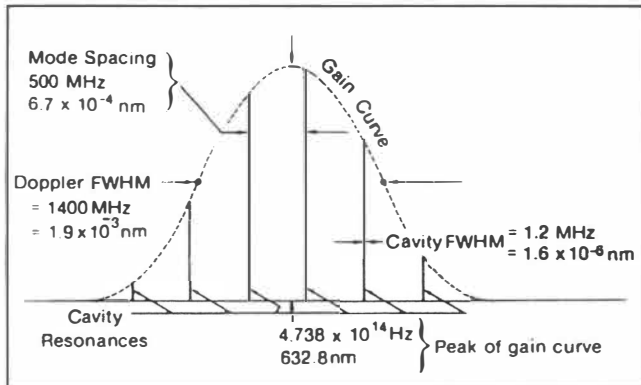


LASER CAVITY TYPES, showing the shape and size of the active plasma volume. The most popular cavity for HeNe lasers is the quasi-hemispherical cavity, because of its stability and short length. To obtain a quasi-hemispherical cavity, the hemispherical cavity is shortened very slightly.

Various shapes of resonant cavity have been used. The plane parallel resonator, perhaps the most obvious choice, is never used. It has the theoretical advantage of fully utilizing the available plasma volume to obtain high efficiency, but is inherently unstable and has large diffraction losses.

The confocal resonator has two equal radius concave spherical mirrors, each placed at the center of curvature of the other. The cavity utilizes a large fraction of the plasma volume and produces high power. It is less sensitive to alignment, but is slightly sensitive to mirror separation. The confocal resonator is rarely used when TEM₀₀ output is required.

The hemispherical resonator has a plane mirror at the center of curvature of a concave spherical mirror. The resonator becomes quasi-hemispherical if this mirror separation is slightly reduced. This quasi-hemispherical configuration is easy to adjust and produces highly coherent output. The disadvantage of this cavity is that it only utilizes about one-third the volume of available plasma due to the focusing effect of the concave mirror. This inefficiency limits the power output. However, its ease of adjustment, and stability once aligned more than outweigh the disadvantage of power limitation. Quasi-hemispherical cavities are, therefore, the most commonly used in commercial lasers.



DOPPLER BROADENED GAIN CURVE supporting 6 longitudinal modes or cavity resonances. The gain curve is of the same shape as the line profile in spontaneous emission.

The presence of an optical cavity leads to longitudinal modes of oscillation. Only the light wavelengths which correspond to standing waves are amplified. Out of phase reflections are quickly lost through destructive interference. In order for a mode or standing wave to exist, the round trip cavity length must be exactly equal to an integer number of wavelengths:

$$2L = m\lambda \text{ (where } m \text{ is a large integer).}$$

This corresponds to the series of resonant frequencies

$$\nu = \frac{mc}{2L} \text{ (where } c = \text{velocity of light).}$$

These frequencies each define an extremely narrow band within which laser oscillation can occur. The frequency separation of these longitudinal modes or resonances is

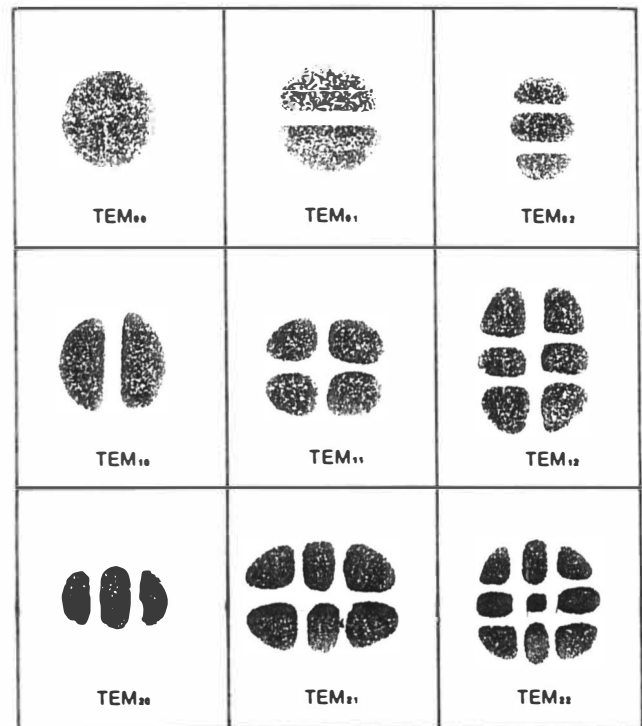
$$\Delta\nu = \frac{c}{2L},$$

and is inversely proportional to the length of the cavity. Thus, for a 50cm cavity length, the mode spacing will be:

$$\begin{aligned} \Delta\nu &= \frac{3 \times 10^{10}}{2 \times 50} \text{ Hz} \\ &= 300 \text{ MHz} \end{aligned}$$

In order for these modes to exhibit optical gain, their individual frequencies must lie within the finite spectral line width of the natural atomic transition. The net effect of random thermal atomic motion within the plasma is to produce a Doppler broadened spectral line. Modes which fall within the Doppler broadened spectral envelope will be amplified. The linewidth of a single laser mode is extremely narrow compared with the Doppler linewidths ordinarily encountered in atomic-spectroscopy.

In addition to longitudinal modes, the cavity can support a number of Transverse Electric and Magnetic modes (TEM)



TRANSVERSE ELECTRIC AND MAGNETIC MODES of a laser cavity, with identification, and showing nulls of the irradiance distribution. The solid regions represent the interiors of irradiance contours arbitrarily selected to best reveal distribution shape.

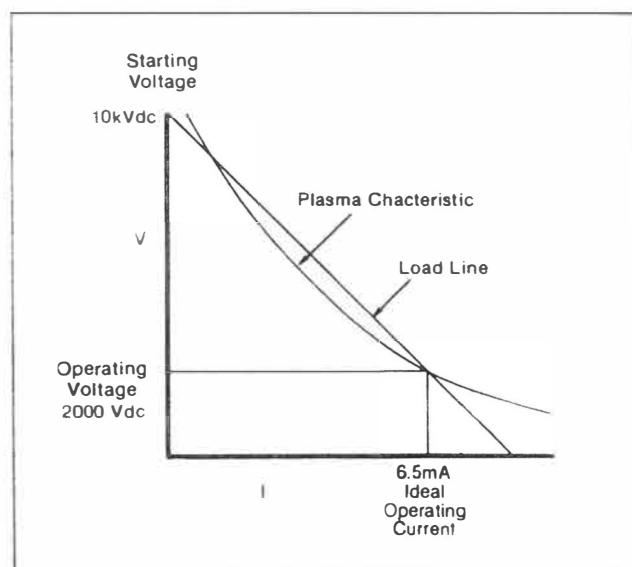
modes). These are modes in which the irradiance distribution along the electric and magnetic field vectors exhibits one or more nulls. At small angles from the cavity axis it is possible to obtain a different series of standing waves, each of which can produce a stable mode of oscillation and gain. Transverse modes are identified by their irradiance distributions and are designated TEM_{pq} , where the subscripts p and q refer to the number of nulls along two orthogonal axes which are perpendicular to the cavity axis. The lowest order TEM_{00} mode has a Gaussian irradiance distribution. This mode experiences the minimum possible diffraction loss, has minimum divergence, and can be focused to the smallest possible spot. For these reasons, it is often desirable that the laser should be restricted to operation in this fundamental mode. Higher order modes have a larger spread and usually suffer high diffraction losses. A mode often encountered in practice is the "doughnut" mode (TEM_{01}), which is a combination of TEM_{01} and TEM_{10} , having a central minimum and a bright ring.

Lasers are sometimes designed to exclude all but the fundamental mode, and sometimes to permit higher order modes. The exact design requirements depend on the type of cavity in use. If higher orders are to be excluded, the bore size must be held very close to the fundamental mode diameter. This may result in diffraction of some wanted radiation out of the beam. With a confocal resonator it is very difficult to exclude high order modes. Most TEM_{00} mode lasers use quasihemispherical resonators. In this case, beam collimation quality is limited only by diffraction. Most helium neon lasers have an output beam divergence of 0.8 to 1.5 milliradians (beam angular diameter).

The polarization of laser radiation is an important effect which occurs as a direct result of the stimulated emission process. Just as the stimulated photon is in phase, frequency, and directional agreement with the stimulating photon, so it also carries the same state of polarization. The resonant cavity sets up standing waves of constant linear polarization. This results in alternate longitudinal modes having orthogonal states of polarization. Likewise, adjacent transverse modes have orthogonal states of polarization. The resultant polariza-

tion depends upon the precise mix of modes present, and this mix may change rapidly with time. So called "randomly polarized" lasers have an indeterminate combination of orthogonally polarized radiation. To eliminate this uncertainty, laser outputs are often deliberately linearly polarized. A single Brewster angle window is placed in the cavity, effectively eliminating one state of polarization, and preventing it from being amplified. The output is more stable and has a known state of polarization, namely the p-polarization defined by the window.

Electrically, helium neon lasers function as negative resistors. They require a high voltage in order to strike the discharge and then as current flows the voltage required to sustain operation is reduced. A ballast resistor is connected in series with the laser to provide safe minimum impedance. The output of the laser is a function of the d.c. discharge current. In order to optimize the laser performance, a particular operating current is chosen, and the power supply is designed to maintain this current with very low ripple.



CHARACTERISTIC OPERATING CURVE OF A HeNe LASER. The curve is typical of other low pressure gaseous discharges, such as advertising signs.

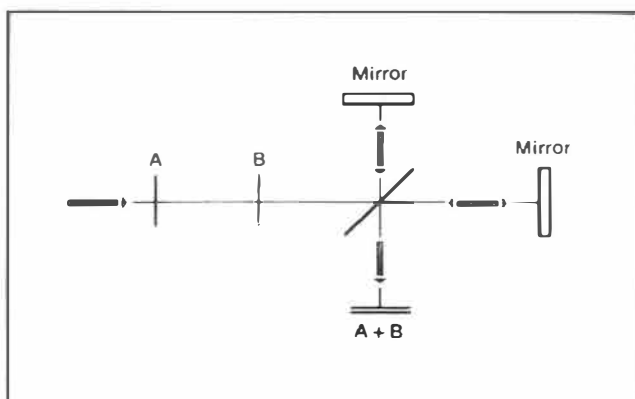
CHARACTERISTICS OF LASER LIGHT

The helium neon laser delivers relatively intense coherent electromagnetic radiation at 632.8nm. It is able to do this reliably, for many hundreds of hours, without variation of output power or beam direction, at modest cost. The temporal coherence of a laser is a measure of the ability of the beam to produce interference effects as a result of differences in path lengths, and is important in interferometry and holography. The spatial coherence is of importance in applications where it is necessary to focus all of the laser's output into an extremely small spot.

Both temporal and spatial coherence have long been sought for industrial and laboratory optical applications. Before the laser was invented, monochromatic sources with various degrees of coherence were in use. The relatively low output levels available, and the Doppler broadening of most atomic sources, were such that only limited applications were ever achieved.

TEMPORAL COHERENCE

If the phase at a particular instant in time along a traveling



MICHELSON INTERFEROMETER, with path difference adjusted so as to superimpose wavefronts A and B in the output beam, where they will interfere if their separation in the input beam did not exceed the radiation coherence length. The interferometer measures coherence length, and coherence length is the measure of temporal coherence.

wavefront is the same as its phase after the wave has traveled a distance x in x/c seconds, for all values of x , the wave is defined to be perfectly temporally coherent. Michelson interference fringes are a direct result of temporal coherence, and they are visible only over the distance in which the temporal coherence is maintained. For conventional sources this distance is rarely more than a few millimeters, whereas for a multimode HeNe laser it is typically 20 to 30 centimeters. That this distance is not infinite demonstrates that lasers are not perfectly monochromatic or perfectly coherent. Off-the-shelf lasers are good enough, however, to permit precise interferometric measurement of distances hundreds of times longer than was formerly possible.

Visibility of interference fringes is defined as

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$$

where $I_{\max}$ and $I_{\min}$ are the maximum and minimum observed fringe intensities. Unity visibility is associated with full temporal coherence. Zero visibility (absence of fringes) corresponds to $I_{\max} = I_{\min}$, and is associated with complete incoherence.

The minimum path length difference for which V is zero (i.e., fringes vanish) is called the coherence length (S_c). The time delay corresponding to this length is the coherence time (Δt). These are related by $S_c = c\Delta t$. The uncertainty relation between time and frequency leads to $\Delta\nu \Delta t = 1$, where $\Delta\nu$ is the line width (frequency spread) of the non-monochromatic laser output line. Lasers operating in a single longitudinal mode have linewidths of approximately 1 MHz. A typical FWHM for a commercial multimode HeNe laser is $\Delta\nu \approx 1500$ MHz, because the Doppler broadened bandwidth provides an envelope for the gain of a number of longitudinal modes.

SPATIAL COHERENCE

Correlation of phase in light beams transverse to the direction of travel is best illustrated by Young-type interference fringes. The light from two transverse portions of the

same beam is allowed to interfere, producing fringes. Visibility is defined as above. Visibility of unity will again correspond to complete coherence, whereas visibility of less than unity implies partial coherence.

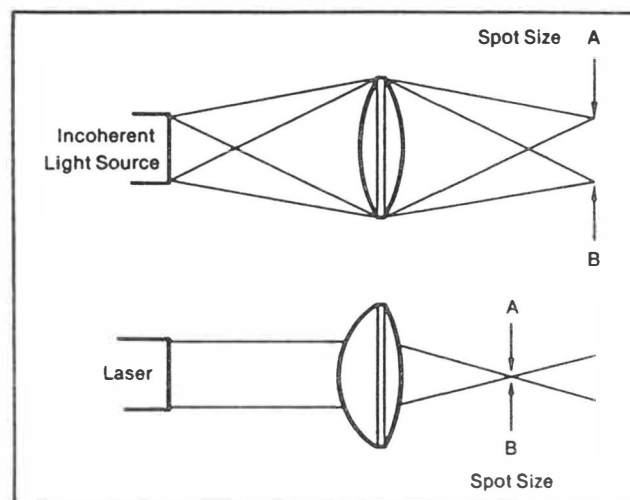
Spatial and temporal coherence are independent of one another; a source may be spatially coherent while temporally incoherent and vice versa. The mode discrimination properties of the laser cavity produce spatial coherence, and the Doppler broadened bandwidth limits temporal coherence.

DIRECTIONALITY AND ENERGY DENSITY

The extreme directionality of laser radiation follows directly from the coherence of the stimulated emission process. Because all of the stimulated photons are alike in every way, they carry no information about the location of the excited neon atoms they come from. Thus all of the excited neon atoms are functionally equivalent, and the laser output radiation behaves as if it all had originated in a tiny imaginary volume, having dimensions of the order of a wavelength, situated precisely on-axis at the center of the cavity waist. This remains true regardless of the size or shape of the active plasma volume.

When an off-axis neon atom is stimulated to emit a photon, the appearance presented to the outside world is that this photon originates on-axis. This behavior is attributable to transverse coherence. When a neon atom far from the waist plane is stimulated to emit, the photon still appears to come from the waist plane. This behavior is attributable to temporal coherence far in excess of the monochromaticity requirements of the beam optics formulas. Because all the coherent energy output of the active medium is effectively funneled by stimulated emission through that tiny imaginary volume, energy densities in focused laser beams can far exceed the highest energy densities actually occurring in the source plasma, a situation impossible with spontaneous emission.

When we look at an extended incoherent source, such as the surface of an incandescent solid or the spontaneous emission of a gaseous discharge, we can plainly see that it is extended.



FOCUSABILITY COMPARISON for light from laser and conventional sources. Extended sources have extended images. The laser is the functional equivalent of a true point source situated at the center of the cavity waist.

This is because the photons carry directional information telling us which parts of the source they come from. Laser radiation carries no such information.

SPECKLE

A smooth but diffuse reflector has a surprising appearance under HeNe laser illumination. Though the beam irradiance distribution may be very smooth, the apparent distribution at the surface is granular. The granule size is approximately the size of the resolution element of the eye. Neighboring granules vary enormously in brightness. A bright granule looks bright because of cooperative in-phase reflections from two or more area elements within the granule, directed at the eye. Such interference is possible because coherence length exceeds the scale of surface roughness. Speckle is a great nuisance in laser imaging and holography, but carries information valuable in non-contacting metrology. Radar "glint" is the same effect on a larger scale, and hampers radar imagery.



Making use of a laser often involves the use of lenses and other optical elements to modify or shape the laser beam.

In some respects laser component design and tolerancing are more critical than for conventional optical components. The designs tend to be simpler, however, because many of the constraints associated with imaging systems are not present. Laser beams are nearly always used on axis, eliminating the need to design for asymmetric aberration correction. Chromatic aberrations are of no concern in helium neon lasers, although they are critical for some dye laser applications. In fact the only significant aberration in most helium neon applications is primary (3rd order) spherical aberration. Focal ratios are often very high, making even this aberration insignificant.

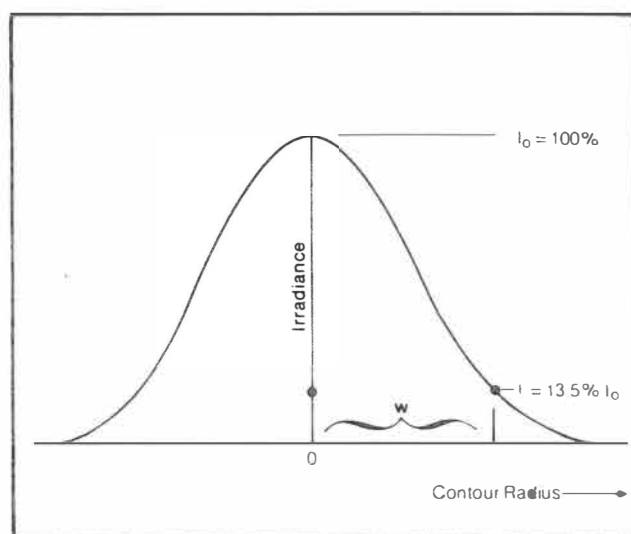
Scatter from surface defects, inclusions, dust, or damaged coating gives more concern than in incoherent systems. Speckle content arising from object texture and radiation coherence can limit system performance.

Because laser light is generated coherently over a relatively large volume, it is not subject to some of the limitations normally associated with incoherent sources. All parts of the wavefront act as if they originated from the same point, and consequently the emergent wavefront may be precisely defined. Starting out with a well-defined wavefront permits more precise focusing and control of the beam than would otherwise be possible. In laser component design both aberration and diffraction effects must be considered.

In order to gain an appreciation of the principles and limitations of beam optics it is necessary to understand the nature of the laser output beam. In the TEM_{00} mode the beam emitted from a laser has a perfect plane wavefront and a Gaussian transverse irradiance profile. The Gaussian shape is truncated at some diameter either by the laser bore or some limiting aperture in the optical train. To specify and discuss the propagation characteristics of a laser beam we must define its diameter in some way. The commonly adopted definition is the diameter at which the transverse field amplitude has fallen to a fraction $1/e$ of its peak axial value. At this same diameter the corresponding beam irradiance will have fallen to $1/e^2$

(13.5%) of its peak or axial value.

A Gaussian beam can be collected and focused to a small area even at large distances from the laser, but all optics must approach diffraction limited performance.



GAUSSIAN IRRADIANCE PROFILE for TEM_{00} mode, showing definition of beam radius w .

BEAM DIVERGENCE

Physical law dictates that waves must spread transversely as they propagate, and it is therefore impossible to have a perfectly collimated beam. The spreading of HeNe laser beams is in precise accord with the predictions of pure diffraction theory; aberration is totally insignificant in the present context. Under quite ordinary circumstances the spreading can be so slight as to be very difficult to see, and many people do not realize that the spreading occurs at all. The formulas which follow accurately describe beam spreading. Through them it is easy to see the capabilities and limitations of laser beams. The notation is consistent with much of the laser literature, and in particular with Siegman's excellent *Introduction to Lasers And Masers* (McGraw-Hill, pp. 304-317).

Even if a Gaussian TEM₀₀ laser beam wavefront were made perfectly flat at some plane, with all elements moving in precisely parallel directions, it would quickly acquire curvature and begin spreading in accordance with

$$R(z) = z \left[1 + \left(\frac{\pi w_0^2}{\lambda z} \right)^2 \right]$$

and

$$w(z) = w_0 \left[1 + \left(\frac{\lambda z}{\pi w_0^2} \right)^2 \right]^{1/2}$$

where z is the distance propagated from the plane where the wavefront was flat, λ is the wavelength of light, w_0 is the radius of the $1/e^2$ irradiance contour at the plane where the wavefront was flat, $w(z)$ is the radius of the $1/e^2$ contour after the wave has propagated a distance z , and $R(z)$ is the wavefront radius of curvature after propagating a distance z . $R(z)$ is infinite at $z = 0$, passes through a minimum at some finite z , and rises again toward infinity as z is further increased, asymptotically approaching the value of z itself. The wavefront shape and irradiance distribution changes are precisely analogous to the changes between the near-field and far-field patterns of a dipole antenna. These changes occur without violation of the speed of light constancy postulate of Special Relativity, because of the distinction between group and phase velocity. The plane $z = 0$ marks the location of a Gaussian "waist," or a place where the wavefront is flat, and w_0 is called the waist radius. A waist occurs naturally at the mid-plane of the symmetric confocal cavity. Another waist occurs at the surface of the planar mirror of the quasi-hemispherical cavity used in Melles Griot lasers.

The normalized irradiance distribution of the Gaussian TEM₀₀ beam, namely

$$I(r) = I_0 e^{-2r^2/w^2} = \frac{2}{\pi w^2} e^{-2r^2/w^2}$$

is the same at all cross sections of the beam, where $w = w(z)$. The invariance of the form of the distribution is a special consequence of the presumed Gaussian distribution at

$z = 0$. If instead a uniform irradiance distribution had been presumed at $z = 0$, the pattern at $z = \infty$ would have been the familiar Airy disc pattern given by a Bessel function, while the pattern at intermediate z -values would have been enormously complicated. See Born & Wolf, *Principles of Optics*, 2nd edition, Pergamon/Macmillan, pages 396 and 440.

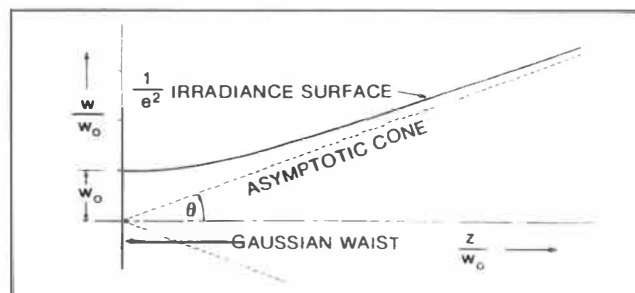
Simultaneously as $R(z)$ asymptotically approaches z for large z , $w(z)$ asymptotically approaches the value

$$w(z) \cong \frac{\lambda z}{\pi w_0}$$

where z is presumed much larger than $\frac{\pi w_0^2}{\lambda}$, so that the $1/e^2$ irradiance contours asymptotically approach a cone of angular radius

$$\theta = \frac{w(z)}{z} = \frac{\lambda}{\pi w_0}$$

This value is the far field angular radius of the Gaussian TEM₀₀ beam. The vertex of the cone lies at the center of the waist.



GROWTH IN $1/e^2$ CONTOUR RADIUS with distance propagated away from Gaussian waist. The effect is due to pure diffraction.

At every z -value, including ∞ , an irradiance contour of radius 1.5 times larger than $w(z)$ is, to an excellent approximation, the 99% throughput contour (meaning that 99% of the beam power passes through it). Corresponding to the cone of angular radius θ mentioned above is a larger cone of angular radius 1.5θ and sharing the same vertex. Unless beam energy losses of 1% or more (along with other undesirable effects) are

acceptable, all optical elements through which it is intended that the beam should pass must be large enough to accept this larger cone. It is a value 1.5θ or larger, and not θ itself, which should be identified with f-number requirements for these elements.

The spectrally selective coating of the spherical output mirror is actually supported on the concave surface of a weak spherical positive meniscus lens. This lens operates on-axis at very high f-number, and so introduces no spherical or other aberration. Its power has been chosen to focus the output beam very slightly, just enough to form a new and much larger waist (flat wavefront) near the lens exit pupil. This improves (reduces) the beam divergence θ by a large factor, and correspondingly increases beam usefulness. Our published $1/e^2$ beam diameters are for the $1/e^2$ irradiance contour of this large output waist.

As an example, the Melles Griot 4 milliwatt laser 05 LHR 141 has a published beam diameter of $2w_0 = 0.8\text{mm}$. Accordingly,

$$\theta = \frac{\lambda}{\pi w_0} = \frac{6328 \times 10^{-7}}{(\pi)(0.4)} = 5.04 \times 10^{-4} \text{ radians.}$$

At a distance z of 100 meters

$$\begin{aligned} w(z) &= z\theta \\ &= (10^5)(5.04 \times 10^{-4}) \\ &= 50.4\text{mm,} \end{aligned}$$

which is about 126 times larger than w_0 .

SPATIAL FILTERING

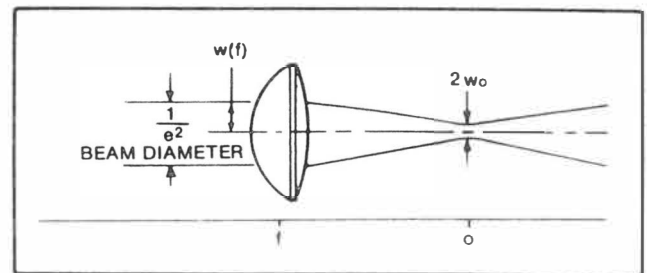
Laser light scattered from dust particles residing on optical surfaces may interfere with the direct laser light to produce interference patterns resembling Fresnel zone plates. Such patterns are very bothersome in interferometric and holographic applications, where they constitute a highly detailed, contrasty, and confusing background from which desired information must be salvaged. Spatial filtering is a simple way of suppressing this background and maintaining very smooth beam irradiance distributions. Spatial filtering depends on the fact that the scattered light propagates in directions different from the direct laser light, and hence arrives at a lens focal plane at places different from that where the direct beam is focused. By centering a small aperture around the focal spot of the direct beam, it is possible to block the scattered light while allowing the direct beam to pass unscathed. The result is a fan or cone of light with a very smooth irradiance distribution, and which may be refocused to form a collimated beam that is almost equally smooth.

To find the size of the focal spot formed when a perfectly collimated laser beam is focused with an aberration free lens (such as a Melles Griot LAO or new LAU series achromat), we may again apply the asymptotic formula

$$w(z) = \frac{\lambda z}{\pi w_0}$$

In the present problem the distance z becomes the lens focal length, f , $w(z) = w(f) = w$ is the $1/e^2$ contour radius at the lens, and w_0 is the contour radius of the new and very small waist formed near the lens focal point. Thus

$$w_0 = \frac{\lambda f}{\pi w}$$



CONCENTRATION OF LASER BEAM by a laser line focusing singlet. Size of the focal waist has been greatly exaggerated for illustrative purposes.

For ease of alignment it is best that the aperture diameter be about 3 times the $1/e^2$ beam contour diameter at the focus, or about 2 times the 99% throughput contour diameter. Spatial filters usually consist of a microscope objective and pinhole, with relative position controlled by a fine xyz motion.

For a microscope objective of focal length f , the optimum pinhole diameter may be taken as

$$d = \pi w_0 = \frac{\lambda f}{w}$$

A pinhole of this diameter centred at the focus will allow 99.3% of the beam power through.

remarkable diffraction effects are observed on ordinary surfaces, indeed on virtually any surface. This 'granular effect' is particularly well observed with a visible gas laser. It needs coherent light, otherwise on any element there is a time-averaging effect which tends to make all the elements look alike. The surface then appears matt, which is one's normal experience with ordinary light.

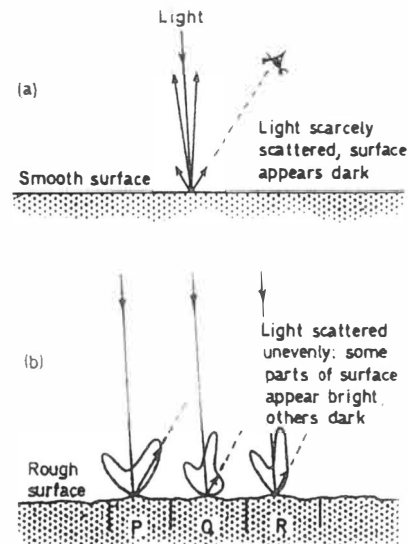


Fig. 15.2. Light is scarcely scattered by a smooth surface (a), but each element of the rough surface (b) scatters differently

Apparently the granular effect was quite unexpected by the inventors of the visible gas laser and took them by surprise.¹ Different observers saw the effect differently, some seeing it as dark spots on a light background and some as bright spots on a dark background.

The effect can be photographed, and Fig. 15.1 shows a photograph of the Brewster-angle window and the external mirror of a gas laser. The patches of light are due to scattering, since the main part of the light is going through the window and

the mirror. The granulations can be seen in the two patches of light.

Several authors have discussed this effect,^{1,2,3,4,5} but it seems true to say that it has not received a systematic experimental investigation.

Consider Fig. 15.2 (a) first, in which light which is falling normally on a very smooth surface is reflected only in one

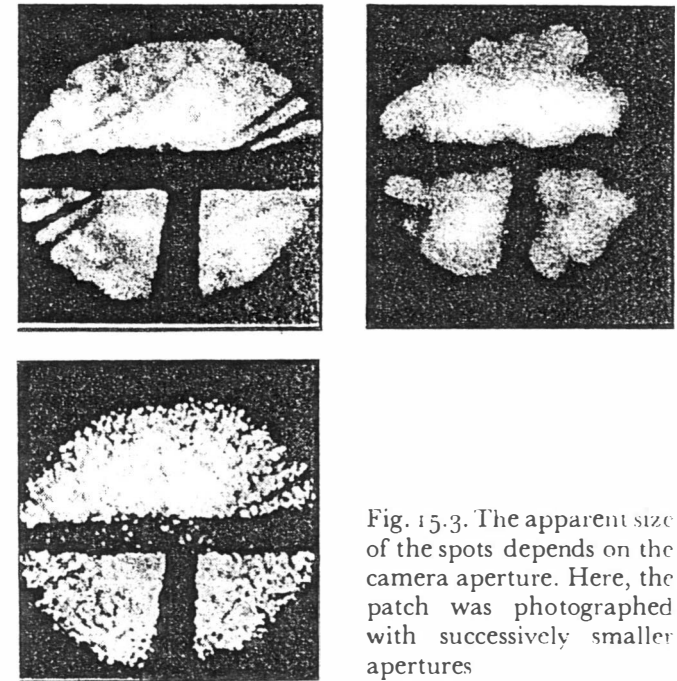


Fig. 15.3. The apparent size of the spots depends on the camera aperture. Here, the patch was photographed with successively smaller apertures

direction, with very little scattering. If the observer is looking from one side, as in Fig. 15.2 (a) the surface appears uniformly dark.

The situation is very different when the surface is rough. Fig. 15.2 (b) shows a rough surface, illuminated by a coherent plane wave. Now, each element of the surface will scatter the light in a rather complicated way which depends on the detail

LASER LIGHT

*Fundamentals and
Optical Communication*

Charles C. Eaglesfield

of its roughness and, generally speaking, the polar diagram of each part of the surface is different. According to how strong the scattering is in the direction of view, the element may appear bright or dark. Thus, in this case, P is bright, Q is rather dim and R is dark — hence the granular appearance.

The size of the granulations depends on the size of the elements that the eye (or the camera lens) can resolve; successive

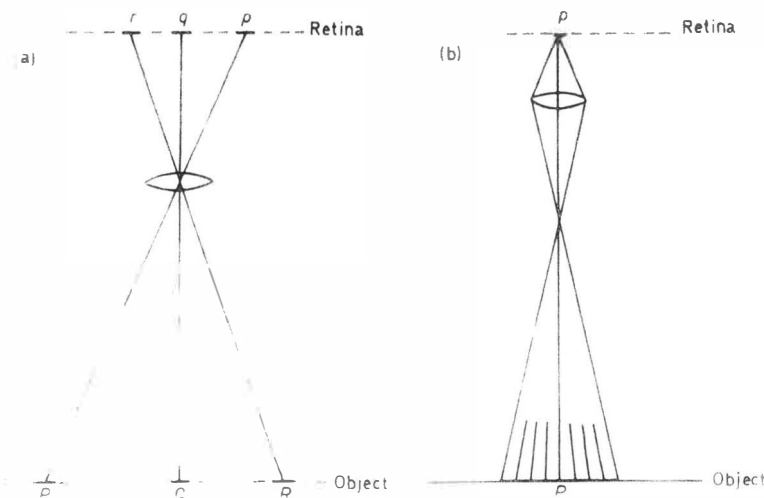


Fig. 15.4. Spots on the retina arise from elements of the surface. The elements are small if the eye is focused on the surface (a) but large otherwise (b).

photographs of the same surface illuminated by laser light, taken with decreasing camera aperture, and hence diminishing resolution, show bigger and bigger granulations. This is shown in Fig. 15.3.

Let us now consider a little more carefully what happens when one views a surface by eye. In Fig. 15.2 the direction of view is shown as being the same for each element. In fact it is not the same, as is shown in Fig. 15.4 (a). The elements of the surface PQR become pqr on the retina, with a slightly different angle of view for each. However, whether the element is bright

or dark still depends on how the angle of view tallies with the polar diagram. The surface still appears granulated.

When the eye is not focused on the surface (Fig. 15.4 (b)) the light reaching p can start from anywhere in the area P , but of course the light from each element of P has to be going in the right direction. Clearly the areas P , Q , etc., overlap, but this is not shown in the diagram, to avoid confusion.

There is rather a strange visual effect associated with defocusing. If one moves one's head slowly a small distance the granulations appear to move across the object. It results from parallax between the object plane and the focus plane. The movement can be in either direction, depending on whether the eye is focused above or below the object. It is not very easy to focus one's eyes on the surface because one's eye tends to 'chase' the spots, so that this relative movement is rather commonly seen.

These visual effects have been described as though they were novel and only recently found with laser light. In fact, it seems they were observed many years ago.

It was explained in an earlier chapter that interference effects are found with ordinary light as long as it is well collimated. The degree of collimation necessary is greater for wideband sources, but it is said that granulations can be observed even with sunlight. It seems to be necessary that the polar diagrams of the surface elements, indicated in Fig. 15.2 (b), should be the same for all wavelengths in the source; this may not be so for all surfaces, however, and it seems possible that not all surfaces appear granulated in collimated sunlight.

The very narrow bandwidth and, of course, its intensity make laser light ideal for observing granulations, and the reader with access to a visible gas laser will find it fascinating to look at, particularly with an eyeglass or a low-power microscope. A particularly interesting effect should be obtained with several lasers of different colours. If their beams are inclined to each other the bright spots of each colour would tend to come

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It has been noted in a number of places,¹⁻³ in general not easily accessible to physics teachers, that in unpolarized (or as they are sometimes called randomly polarized) integral-mirror He-Ne lasers, the output power, when externally polarized, shows large fluctuations, in some cases approaching 100% of total usable output. Those fluctuations result from longitudinal mode sweeping caused by thermal changes in the length of the cavity, alternate longitudinal modes being orthogonally polarized.³ It is not as well known that similar fluctuations arising through the same mechanism occur in some polarized lasers as well.

Integral-mirror He-Ne lasers of the type normally used in teaching laboratories are polarized using the magnetic fields produced by permanent magnets placed around the cavity.³ These lasers have the advantage that they can easily be changed from polarized to unpolarized, and vice versa, by removing or replacing the magnet assemblies in the laser heads. A typical specification is a cross-polarized component at a power less than 3% of the total power (i.e., an extinction ratio of about 30:1 compared to a typical ratio of

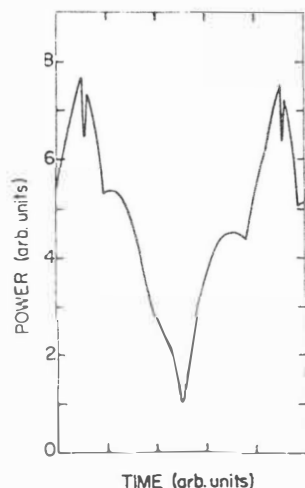


Fig. 1. Fluctuation of the cross component of the output in a polarized 2-mW He-Ne laser. In this case, the period of the fluctuation was about 20 sec.

in undergraduate laboratories.

A fluctuation of the type observed in all polarized units is shown in Fig. 1, where the power of the cross-polarized component of the output is plotted as a function of time. The fluctuations are repetitive, the period depending on the ambient conditions (i.e., temperature and air movement), the time elapsed since turning on and the particular laser model in question. In this case the peak-to-peak change amounted to 10% of the total rated output, with the average power of the cross component at its rated 3% of the total output power. Figure 2 shows the relative magnitude of the fluctuations given by the same laser in series with a linear polarizer as a function of the orientation θ of the polarizer, the angle $\theta = 0$ being the orientation for maximum transmitted average power. Large variations (of the type shown in Fig. 1) occur near the angles $\theta = \pm 90^\circ$, these being the fluctuations one might observe in a Fresnel reflection experiment near the Brewster angle, making meaningful quantitative measurements difficult to obtain. The plot is not symmetric about $\theta = 0$, possibly because the magnetic field of the permanent magnets was not perfectly aligned with, or orthogonal to, the polarizations of successive longitudinal modes. This also explains the complex shape of the fluctuation as shown in Fig. 1.

Fluctuations in the polarization due to mode sweeping cannot be eliminated easily even in the limit of long warming times because of the small length changes ($\Delta L = \lambda/2$) needed to sweep longitudinal modes. For a typical He-Ne laser ($L \approx 25$ cm, $\lambda = 632.8$ nm) this length change corresponds to a temperature change of less than 0.1°C .

500:1 for designs incorporating Brewster windows). While an extinction ratio of 30:1 may seem adequate for many experiments illustrating polarization-dependent effects such as Fresnel reflection, Malus's law, or even diffraction,⁴ serious problems often occur when using these lasers, because of the large mode-sweeping contribution of the cross component. As a result of mode sweeping the power of the cross component fluctuates with periods typically ranging from a few seconds to about a minute, the shortest periods occurring immediately after the laser has been turned on. Many experiments commonly used in teaching laboratories, including the ones mentioned above, are particularly sensitive to fluctuations with periods in this range.

The fluctuations can be illustrated most easily using a simple experimental arrangement consisting of a laser in series with a linear polarizer and a photodetector coupled to an amplifier and a chart recorder. We have investigated a total of eight He-Ne integral-mirror lasers from three different manufacturers, ranging in rated output power from 1 to 5 mW, all of them of the variety commonly used

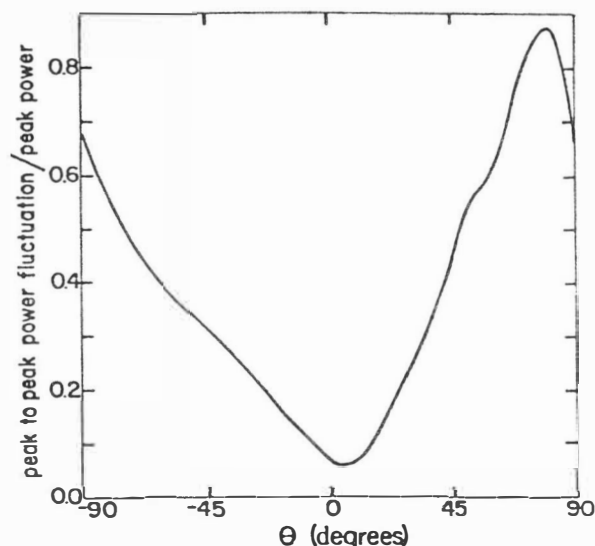


Fig. 2. Normalized power variations given by a 2-mW polarized He-Ne laser in series with a linear polarizer, as a function of the orientation of the transmission axis of the polarizer.

The use of linear polarizers to cut off the cross polarized modes may not be adequate in many experimental situations because the power of the orthogonally polarized modes tend to fluctuate in such a way as to keep the total output power approximately constant. Thus with the cross polarized modes cut off by a linear polarizer, the available output power may fluctuate sufficiently to cause serious difficulties. However, with an unpolarized laser, it is possible to obtain a stable polarized output by using a linear polarizer oriented at 45° with respect to the polarizations of the modes as discussed by Mas³ *et al.*

Thus in experiments involving polarizing elements and quantitative intensity measurements He-Ne lasers polarized using permanent magnets may not be suitable, not so much because of the smaller extinction ratios these lasers offer compared with Brewster window types, but because of the nature of the fluctuations in polarization that occur.

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